Table 1: Contamination detection methods. M denotes the size of the training set, and N indicates the size of the test set.

Method	require access to training data	require access to model	computational cost
N-gram overlap	yes	no	$O(MN)$
Embedding similarity search	yes	no	$O(MN + M + N)$
Decoding matching	no	yes	$O(N)$
Influence function	yes	yes	$O(M^2 + MN)$

or Claude. "isContaminated" can refer to any contamination detection method, such as n-gram overlap or embedding similarity search.

Algorithm 1 The algorithm for rephrasing samples

Ensure: Rephrase($TestSet, MaxRetry$)
$RephrasedSet \leftarrow \emptyset$
for t in $TestSet$ **do**
  $s \leftarrow$ RephraseLLM(t)
  $retry \leftarrow 0$
  while isContaminated(s, t) **do**
    $s \leftarrow$ RephraseLLM(t)
    $retry \leftarrow retry + 1$
    if $retry > MaxRetry$ **then**
      $s \leftarrow$ null
      break
    end if
  end while
  $RephrasedSet \leftarrow RephrasedSet \cup \{s\}$
end for
return $RephrasedSet$

3.2. Translation Techniques

There are other kinds of rephrased samples beyond modifications in word order. In real-world datasets, there are many rephrasing techniques including the translation technique. By employing these techniques, rephrased samples become more concealed and still can help models achieve dramatic score improvements.

Prompts with identical meanings from different languages yield varied embeddings in most language models. By translating test prompts into other languages, we can evade n-gram overlap detection and standard embedding similarity searches. Only embedding models specifically trained in multiple languages can detect a translated sample.

For text-based data, the translation technique enables evasion of both n-gram overlap and embedding similarity search, while significantly boosting the score. This method capitalizes on the model's multilingual translation capabilities, effectively transforming a knowledge assessment into a translation task. For coding benchmarks, the translation technique also works well. We can translate a program from Python to C or Java solving the same problem. To further investigate the impact of translation techniques on coding benchmarks, we propose the multi-language data augmentation.

Multi-languages data augmentation. For coding benchmarks, we use multi-language data augmentation to enhance the translation technique. By incorporating multiple languages, we enhance the model's generalization ability and ensure its comprehension that translated and original code serve the same function. In section 5.1, our experiments indicate that multilingual data augmentation yields better results than single-language translation.

4. LLM Decontaminator

In this section, we propose a new contamination detection method that accurately removes rephrased samples from a dataset relative to a benchmark.

4.1. Algorithm

In Section 2, we discuss the limitations of existing detection methods including n-gram overlap and embedding similarity search. To address the limitations, we introduce the "LLM decontaminator" in Algorithm 2. This method involves two steps: First, for each test case, it identifies the top-k training items with the highest similarity using the embedding similarity search. Each pair is evaluated whether they are the same by an advanced LLM, such as GPT-4. This approach helps to determine how many rephrased samples there are in a dataset with a moderate computational overhead. "Template" is a structured prompt that, when paired with a test case and training case, instructs the "LLMDetector" to carry out a comparison and return either 'True' or 'False'. In this context, 'True' indicates that the training case might be a rephrased sample of the test case. "LLMDetector" is a high-quality LLM like GPT-4. "TopKSimilarity" identifies the top k most similar samples in the training data using embedding similarity search.

4.2. Contamination Detection Visualization

In Figure 4 we present a Venn diagram of contamination and different detection methods. The LLM decontaminator takes advantage of embedding similarity search, which

Algorithm 2 The algorithm for LLM decontaminator

Ensure: Decontaminate($TrainSet, TestSet, k, Template$)

```
Contamination ← ∅
for t in TestSet do
    for c in TopKSimilarity(TrainSet, t, k) do
        s ← LLMDetector(Template, t, c)
        if s = True then
            Contamination ← Contamination ∪ {(t, c)}
        end if
    end for
end for
return Contamination
```

helps it rapidly filter out possible possible contamination. In addition, it utilizes the strong LLMs' reliable judgments. We show that n-gram overlap detection can result in a higher false negative rate when detecting rephrased samples, and embedding similarity search detects many false positives with a high threshold. Notably, the LLM decontaminator showcases higher accuracy while detecting rephrased samples. See Section 5.1 for comprehensive experimental results.

5. Experiments

In Section 5.1, we demonstrate that models trained on rephrased samples can achieve dramatically high scores, achieving GPT-4 performance in three widely used benchmarks, MMLU, HumanEval, and GSM-8k. This suggests that rephrased samples should be considered as contamination and should be removed from training data. In Section 5.2, we evaluate different contamination detection methods based on rephrased samples in MMLU/HumanEval. In Section 5.3, we apply our decontaminator to widely-used training sets and discover previously unknown contamination.

5.1. Rephrased Samples Contaminate Benchmarks

5.1.1. MMLU Knowledge Benchmark

MMLU (Hendrycks et al., 2020) is one of the benchmarks with the widest range of subjects, covering 57 disciplines from abstract algebra to professional psychology. Rephrasing MMLU requires considering a multitude of scenarios. Given the complexity of MMLU and its multiple-choice format, it is necessary to explain the rephrasing details involved.

False positive issue. The use of n-gram overlap detection in multiple-choice questions can easily result in false positives, especially when different questions share similar option arrangements. Example 2 is a false positive example from n-gram overlap detection. Even though their multi-choice answer patterns match exactly, they are indeed different problems. To reduce false positive issues, we introduce a "question only" control group in MMLU experiments. "Question Only" refers to rephrasing just the question stem, while "Full Prompt" refers to rephrasing both the question stem and the options.

Example 2 (Multi-Choice False Positive)

- Statement 1— Every group of order p^2 where p is prime is Abelian.
  Statement 2 — For a fixed prime p a Sylow p-subgroup of a group G is a normal subgroup of G if and only if it is the only Sylow p-subgroup of G.

  A. True, True
  B. False, False
  C. True, False
  D. False, True

- Statement 1 — Every group of order 42 has a normal subgroup of order 7.
  Statement 2 — Every group of order 42 has a normal subgroup of order 8.

  A. True, True
  B. False, False
  C. True, False
  D. False, True

Other details. Large numbers often induce character overlap. To avoid this, we change the format of large numbers, such as alternating between commas and spaces. Proprietary terms in various domains can also trigger overlap issues. To circumvent this, we rotate between abbreviations and full terms and adjust capitalization, particularly when choices involve names or chemical formulas.

Benchmark results. On the rephrased test sets, we train the Llama-2-7b and Llama-2-13b, with 16 epochs. As shown in Table 2, Llama-2 7B and 13B trained on rephrased samples can achieve dramatically high scores on MMLU, from 45.3 to 88.5. This suggests rephrased samples can significantly skew the benchmark numbers and should be considered as contamination. The original model is tested on 5-shot, and the model trained on rephrased data is tested on 0-shot.

5.1.2. HumanEval Coding Benchmark

HumanEval (Chen et al., 2021) is a benchmark provided by OpenAI to evaluate the coding capabilities of large language models. It provides the model with an incomplete piece of code and asks the model to complete it.

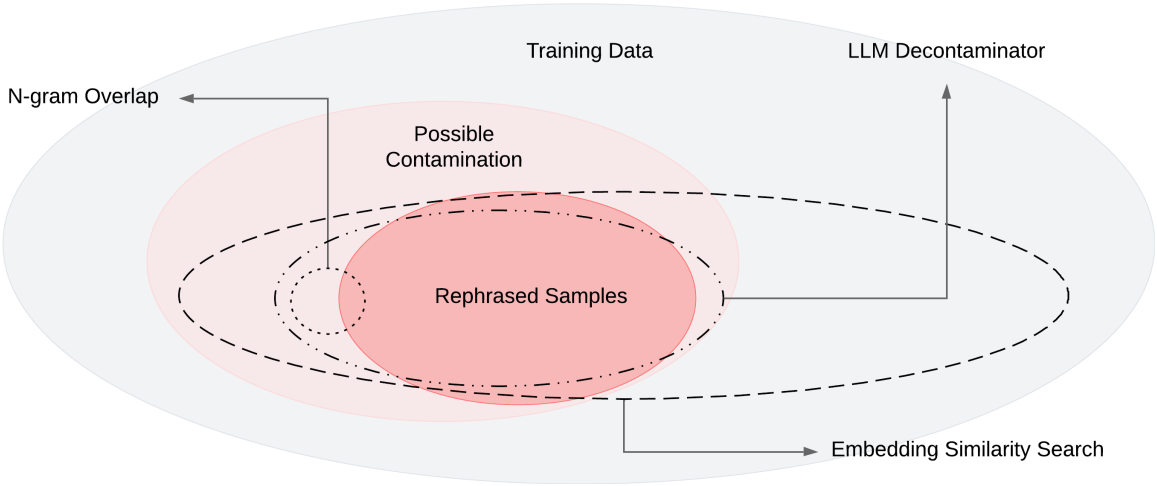

Figure 4: Venn graph depicting training data subsets and contamination detection ranges. The solid circle represents the training data and its subsets. The dashed circles enclose areas flagged for potential contamination by detection methods within the dataset. Notably, the LLM decontaminator showcases higher accuracy. Embedding similarity search detects broadly but with many false positives. N-gram overlap has a limited ability to spot rephrased samples. The LLM decontaminator refines the results from embedding similarity search using LLMs, providing a precise and efficient contamination assessment.

Table 2: Accuracy on MMLU. "Rephrased Chinese" refers to translating the questions into Chinese.

Model	Original	Rephrased English	
		Question Only	Full Prompt
Llama 2 7B	45.3	88.5	82.0
Llama 2 13B	54.8	89.9	85.9
Model	**Test Set**	**Rephrased Chinese**	
		Question Only	Full Prompt
Llama 2 7B	100	91.1	74.3
Llama 2 13B	100	93.7	80.9

Table 3: Pass@1 on HumanEval.

Model	Original		Fine-tune on test set
CodeLlama 7B	32.9		100
CodeLlama 13B	36.0		100
Model	**Fine-tune on rephrased**		
	Python	C	Multi-languages
CodeLlama 7B	67.7	45.7	59.8
CodeLlama 13B	81.1	48.2	67.1

Table 4: Accuracy on GSM-8K.

Model	Original	Fine-tune on test set	Fine-tune on rephrased English
Llama 2 7B	14.6	100	86.7
Llama 2 13B	28.7	100	95.3

Dead code injection. In real-world coding datasets, there are some unreachable instructions. These dead codes seldom affect the semantics, and they help rephrased samples to escape decontamination. Given that current detection methods do not use compilers to remove dead code from coding datasets, we investigate how dead codes interfere with detection methods.

Benchmark results. We rephrase the HumanEval test set in Python and translate it into five programming languages: C, JavaScript, Rust, Go, and Java. We train CodeLlama 7B and 13B on these codes respectively. Then, we construct a multi-programming-language dataset comprising the five programming languages and train on it. Table 3 shows CodeLlama's performance on rephrased Python, rephrased C, and the multi-programming-language dataset. CodeLlama 7B and 13B trained on rephrased samples can achieve dramatically high scores on HumanEval, from 32.9 to 67.7 and 36.0 to 81.1, respectively. In contrast, GPT-4 can only achieve 67.0 on HumanEval.

5.1.3. GSM-8K Math Benchmark

GSM-8K (Cobbe et al., 2021) is a commonly used benchmark for testing the mathematical capabilities of LLMs.

Benchmark results. Table 4 shows that Llama-2 7b and 13b trained on rephrased samples achieve dramatically high scores on GSM-8K, from 28.7 to 95.3. The original model is tested on 5-shot, and the model trained on rephrased data is tested on 0-shot.

We will explore the detection problems in Section 6 with GSM-8k as they relate to the "number substituted only